



CATHOLIC JUNIOR COLLEGE
JC2 PRELIM EXAMINATION
Higher 2

Answers

CANDIDATE
NAME

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CLASS

2T

INDEX
NUMBER

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BIOLOGY

Paper 4 PRACTICAL

9744/04

14th AUGUST 2017

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Write your Index number, name and class on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
TOTAL	

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **20** printed pages and **1** blank page.

[Turn over

1 Investigation into the effect of changing the concentration of an enzyme on enzyme activity.

The biological molecule, U, reacts with water to form aqueous ammonium carbonate. The enzyme urease catalyses this reaction.

Aqueous ammonium carbonate produces ammonium ions. These form an alkaline solution which causes red litmus paper to turn blue. The time taken for red litmus paper to turn blue can be used to monitor the progress of the reaction.

You are required to investigate the effect of enzyme concentration on this reaction.

You are provided with the following:

- 15 cm³ of 10.0% urease solution, **E**, which is an irritant
- 100 cm³ of distilled water, **W**
- 25 cm³ of a solution of the biological molecule, **U**
- Red litmus paper, total length of about 20 cm

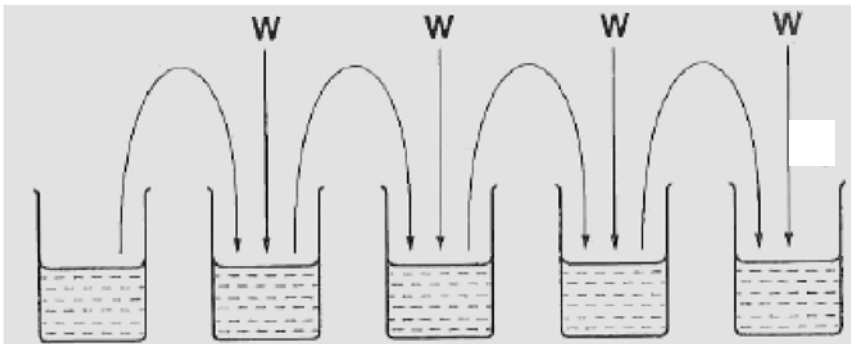
It is recommended that you wear safety goggles / glasses.

- 1 Carry out a serial solution of the urease solution, **E**, to reduce the concentration of the enzyme by half between each of the four successive dilutions, and set up a control.

Label four small beakers, **D1**, **D2**, **D3** and **D4**, for the serial dilutions and label another small beaker, **C**, for the control.

Complete Table 1.1 to show how you will make the different concentrations of urease solution and how you will set up the control, **C**.

Table 1.1

					
Solution	E	D1	D2	D3	D4
concentration of urease / %	10.00	5.00	2.50	1.25	0.650
volume of urease solution to be diluted / cm ³		5.0	5.0	5.0	5.0
volume of distilled water, W / cm ³		5.0	5.0	5.0	5.0
description of the control, C :					
.....					
.....					

Mark Allocation

- [1] – for selecting **10.0, 5.00, 2.50, 1.25 and 0.625** for concentration of urease; expressed consistently in **3 sig. fig.**
 [Accept: if all concentrations are expressed consistently in **3 dec. pl.**]
- [1] – for **correct volumes** in serial dilution, including
- (i) **final volume of each concentration** must be **equal**, i.e. volume of urease and distilled water must add up to give the same final volume
 [Accept: any final volume between 6 to 10 cm³]
 - (ii) **values** are recorded **consistently** and to **appropriate precision**, i.e. **1 dec. pl.**
- [1] – for citing use of distilled water to replace the urease solution for control, **C**; volume of distilled water cited must be consistent with the volume of enzyme / diluted solution.

- 2** In order to monitor the progress of the reaction, in step 4 red litmus paper will be added to each mixture of enzyme (urease) and substrate, **U**, in a test-tube. To prevent the paper sticking to the wall of the test-tube, you will need to use the glass rod to add it, as follows.

Cut a piece of red litmus paper so that it is a little shorter than the circumference of the glass rod. Moisten the paper and stick it to the end of the glass rod as shown in Fig. 1.1. The glass rod can then be lowered into the mixture of urease enzyme and substrate, **U**. The red litmus paper will slip off into the mixture and the glass rod can be removed.

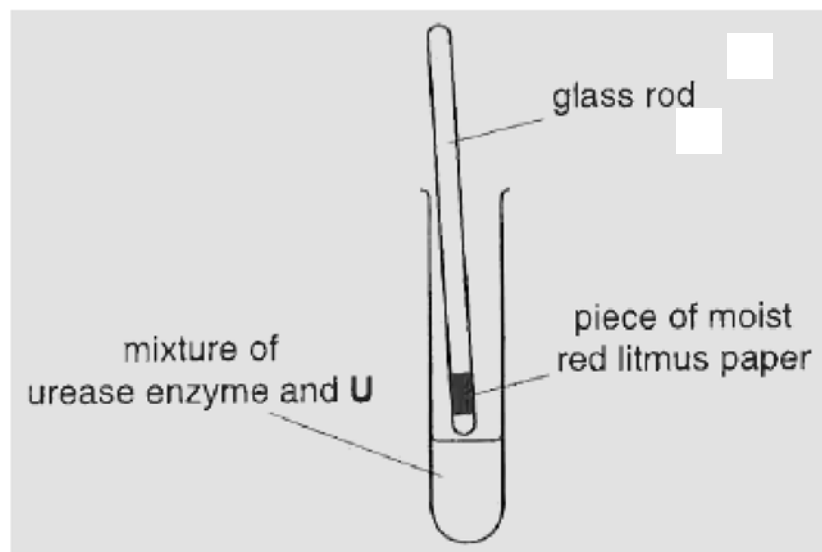


Fig. 1.1

- 3** Prepare a table in the space on page 4 (step 7) to record the results of this investigation at various concentrations of urease solutions, including the control.

Proceed as follows:

- 4** To test the activity of the highest concentration of urease solution, put 2 cm³ of the substrate, **U**, into a test-tube then add 2 cm³ of **E** and mix well. The reaction will start as soon as **E** is added.
 Immediately, put one piece of red litmus paper into the test-tube as described in step 2 and start timing.

- 5 Record, in the table that you have prepared on page 4 (step 7), the time taken for the piece of red litmus paper to turn blue. If the piece of red litmus paper does not turn blue in ten minutes, record 'more than 600'.
- 6 Record steps 4 and 5 for the other concentrations of urease solution, **D1**, **D2**, **D3** and **D4**, and the control, **C**. The red litmus paper used each time should be of the same size.

- 7 Use the space below to record your results.

Concentration of urease solution / %	Time taken for red litmus to turn blue / s
10.0	71
5.00	141
2.50	313
1.25	More than 600
0.625	More than 600
0.00 (control, C)	More than 600

[3]

Mark Allocation

[1] – for results recorded in table format with appropriate column / row heading titles and units

Leftmost column: **Concentration of urease solution / %**

Adjacent column: **Time taken for red litmus paper to turn blue / s**

[1] – for expected trend:

Shortest time for highest concentration of urease solution

Longest time for lowest concentration of urease solution

'More than 600' for Control C

[1] – for values recorded consistently and to appropriate precision, including:

(i) 3 sig. fig. / 3 dec. pl. for urease concentration

(ii) time taken in **whole number**, consistent as per 'more than 600'

[Accept: If time recorded in 2 dec. pl. as per precision of stopwatch used]

- 8 Calculate the rate of reaction, using your result for the 10.0% concentration of the urease solution, E.

For 10.0% concentration of urease solution, E

$$\text{rate of reaction} = 1 / \text{time taken}$$

$$= 1 / 71$$

$$= 0.014 \text{ s}^{-1}$$

rate of reaction0.014 s⁻¹..... [1]

Mark Allocation

[1] – for correct calculation of rate using 1 / time for E / 10.0% urease concentration and units s⁻¹
[Accept: Answer up to 2 sig. fig. / 3 dec. pl.]

- 9 Lack of repeats is one limitation of this procedure. Describe one significant source of error in this procedure that also acts as a limitation.

.....
..... [1]

Mark Allocation

For any **ONE** of the following:

- difficulty in starting the stopwatch at the same time as the start of reaction / delay due to difficulty with inserting litmus paper; OWTTE
- difficulty in judging when the red litmus paper changes colour from red to blue; OWTTE
- AVP

[Reject: for size of litmus paper; temperature; pH or evaporation of water that affects urease concentration or any errors which affects all test-tubes equally]

- 10 Suggest how you would make **one** improvement to this procedure to reduce the effect of the significant source of error identified in step 9.

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.....
..... [1]

Mark Allocation

For any **ONE** of the following:

- Use second person to start time or to add red litmus paper; OWTTE
- Use pH meter / liquid pH indicator with colorimeter / pH sensor with datalogger, in place of the red litmus paper
- AVP

[Reject: for citing use of colorimeter or liquid pH indicator alone]

The effect of pH on the activity of two proteolytic enzymes, **A** and **B**, was compared. The substrate for the enzyme was coloured jelly, which is made of protein.

The apparatus of each pH was set up as shown in Fig. 1.2.

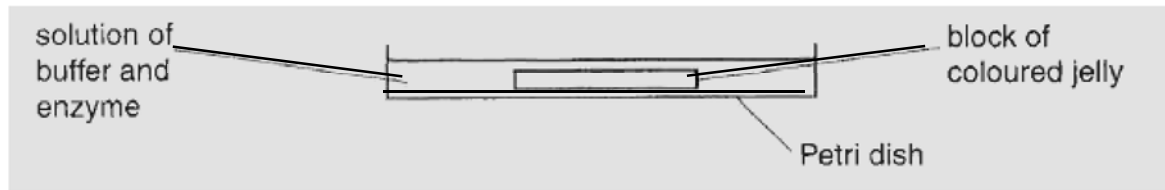


Fig. 1.2

The block of coloured jelly get smaller as it is digested by the enzymes.

- 11** State two variables which would need to be controlled. Suggest how each variable would be controlled.

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..... [3]

Mark Allocation

[1] – for citing **TWO correct variables** from the following:

- (i) **volume** of buffer solution or **volume** of enzymes A and B
- (ii) **concentration** of enzymes A and B
- (iii) **dimension / size** of jelly block
- (iv) **temperature**
- (v) AVP

[2] – for citing **TWO** methods of control that match the 2 identified variables, from the following:

- (i) To use suitable measuring apparatus e.g. syringe, measuring cylinder, graduated pipette - *to transfer / dispense volume*
- (ii) To describe how to make up (e.g. using distilled water to make up to same final volume) – *for enzyme concentration*
- (iii) To measure *jelly block* using ruler or Vernier calipers or grid and cut *to size* using scalpel or knife
- (iv) Use an incubator / thermostatically controlled water-bath, set at a *fixed temperature*
- (v) AVP

The results of the investigation are shown in Table 1.2.

Table 1.2

pH	area of jelly present after 90 minutes / mm ²	
	enzyme A	enzyme B
4.0	10	134
6.4	76	124
7.4	128	76
8.0	138	52
9.0	140	6

- 12** Plot, on the grid opposite, the data shown in Table 1.2. Draw lines of best fit for enzyme **A** and enzyme **B**.
[4]

- 13 (a)** Describe the effect of pH on the activity of enzymes **A** and **B**.

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..... [1]

Mark Allocation

[1] – for citing / description of the effect of (increasing) pH on activity of both enzyme A (decreases) and enzyme B (increases), e.g. reference to comparison of optimum pH of enzyme A and enzyme B)

- (b)** Suggest and explain why changes in pH affect the activity of these two enzymes differently.

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..... [3]

Mark Allocation

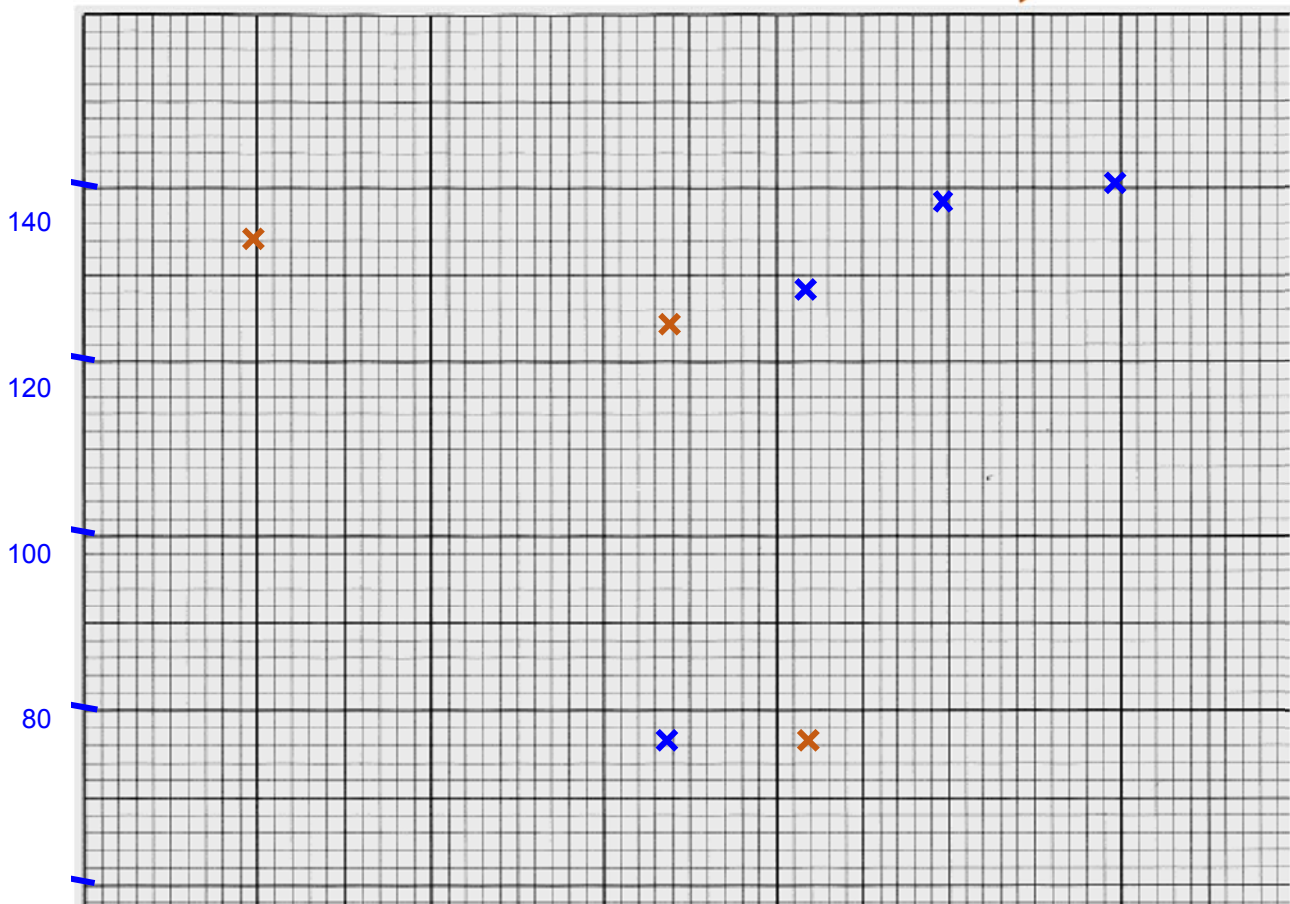
- [1] - any change in pH / concentration of hydrogen ions, changes / affects ionic bonding / ionisation of (acidic / basic) amino acids
- [1] - different amino acids / amino acid side chains, present in enzyme A and B
- [1] – **3D structure / shape / conformation of active site** therefore **changes at different pH values** for enzymes A and B

Area of jelly present after 90 minutes / mm²



Enzyme A

Enzyme B



3.0 4.0 5.0 6.0 7.0 8.0 9.0 pH

Mark Allocation

- [1] – for **appropriate scale** used for **both axes**, with intervals clearly labelled
X-axis: 2 cm (or 10 squares) to 0.5 units interval; $\log_{10}$ 2.0 at origin
Y-axis: 2 cm to 1.0 units; 5.0 ms^{-1} at origin

- [1] – for correct orientation of axes; with appropriate axes titles and units indicated
X-axis: **pH**
Y-axis: **area of jelly present after 90 minutes / mm^2**

- [1] – for correct plotting of **all data points** (as small cross or dot in circle) $\pm$ half a square, for enzymes A and B

- [1] – for data plots joined by **smooth lines of best fit**, both sets of data **distinguished by appropriate labels / key / legend**

CANDIDATE
NAME

CLASS 2T

2 Stomata, in the epidermis of leaves, are responsible for the exchange of gases and the release of water vapour. A pair of guard cells controls the opening and closing of each stoma. In the guard cell membrane there is a transport protein. During the opening of the stomata, this protein uses energy from the hydrolysis of ATP to move protons (H^+) out of guard cells. This has two effects.

- Because protons are positively charged, their removal from the guard cells causes the interior of the cells to become negatively charged relative to the exterior. Because of this, some positive ions such as K^+ move into the interior of the cells lowering the water potential inside the cells.
- The pH inside the cell is increased.

You are provided with leaf samples that are soaking in the following solutions.

solution X: 0.1 mol dm^{-3} potassium chloride at pH 7.0

solution Y: 0.1 mol dm^{-3} sodium chloride at pH 7.0

solution Z: 0.1 mol dm^{-3} potassium chloride at pH 4.5

Proceed as follows:

- 1 Use a pair of forceps to remove the leaf from solution X and use scissors to cut out an area up to $1 \text{ cm} \times 1 \text{ cm}$. Transfer this to a slide ensuring **that the lower epidermis is uppermost**. Use a dropping pipette to add a drop or two of solution X to the leaf surface. Lower a cover slip over the leaf being careful to exclude any air bubbles.
- 2 Use the 10X objective of a microscope to locate the stomata. Count the **total** number of stomata that are visible and the total number that are fully **open** in the same field of view. Ignore any that you are doubtful about. Repeat this for another two areas of the leaf. Calculate the mean percentage of open stomata.
- 3 Repeat steps 1 and 2 for leaves from solutions Y and Z using clean slides and cover slips on each occasion. You **must** keep slide Z in order to answer (b) on page 11.

(a) (i) Record your results in an appropriate format in the space provided below.

Solution	Content	Sample / Area	Total Number of Stomata	Total Number of Open stomata	Mean Percentage of open stomata / %
X	0.1 mol dm^{-3} potassium chloride at pH 7.0	1			
		2			
		3			
Y	0.1 mol dm^{-3} sodium chloride at pH 7.0	1			
		2			
		3			
Z	0.1 mol dm^{-3} potassium chloride at pH 4.5	1			
		2			
		3			

Mark Allocation

^4 Å#hvxow#hfrughg#q#xldeh#deh#rup dw# lk#lssursubwh#hdgljjv##
 ^4 Å#hvxow#hfrughg#ru#6#vdp sdv#frxqw#wrdexpehu#lqg#wrd#shq#wrp dwl,#r#ndfk#r#[/Å#lqg# #
 ^4 Å#fruhfw#ddfxøwrg#r#p hdq#(#ru#shq#wrp dwl
 [1] – uhvxow#krz #n{shfwg#wgg#p ru#wrp dwl#shq#q#[#kdq#q# \#ru# ,

(ii) Explain your results.

X

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Y

[2]

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[2]

Z

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[2]

Mark Allocation [Max 2 for each]

Results with most stomata open (X or Z)

1. (relatively) high concentration of, K^+ / potassium ions, outside cells; K^+ ions moves into guard cells, by facilitated diffusion - reduces water potential
2. water enters by osmosis (down water potential gradient), making (guard) cells turgid - opening stomata

Results in Y (with less opened stomata)

1. (relatively) high concentration of, Na^+ / sodium ions, outside cells but Na^+ enters guard cells slowly / Na^+ does not enter guard cells;
2. Less water enters by osmosis, making (guard) cells less turgid – less opened stomata
3. AVP; e.g. selective channels / channel proteins in guard cell membranes - more for K^+ ions than Na^+ ions; Na^+ ions acts as competitive inhibitor with K^+ for same channel proteins

Result in Z

1. (relatively) high concentration of, H^+ / hydrogen ions, outside cells; - slows down / prevents removal of, H^+ / hydrogen ions, from guard cells;

OR

2. K^+ / potassium ions, outside cells; K^+ ions moves into guard cells, by facilitated diffusion - reduces water potential
3. no $[H^+]$ gradient / H^+ / hydrogen ions, outside cells; - slows down / prevents removal of, H^+ / hydrogen ions, from guard cells;

- (iii) Guard cells, unlike other cells in the epidermis, have chloroplasts. These chloroplasts have grana but lack the enzymes necessary for the light-independent stage of photosynthesis (Calvin cycle).

With results to your results, suggest why guard cells have chloroplasts if they do not carry out the light-independent stage of photosynthesis.

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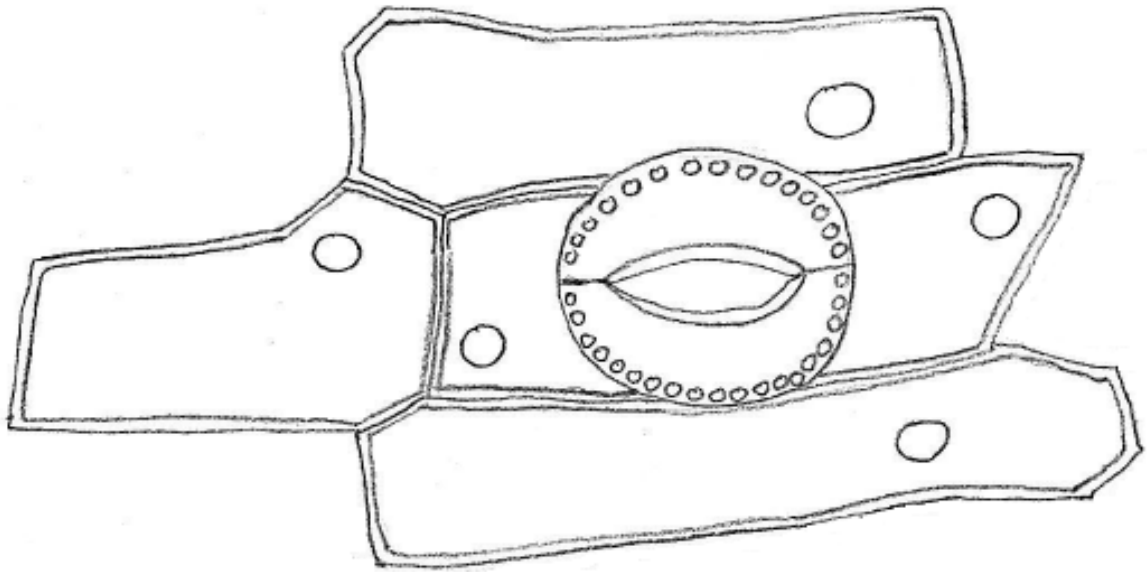
[2]

Mark Allocation

[1] – chloroplasts carry out light-dependent stage, producing ATP (by) photophosphorylation
[Reject: phosphorylation]

[1] – ATP hydrolysed to provides energy for removal of H^+ ions

- (b) Make a **high power** drawing of two guard cells and the epidermal cells on either side of each guard cell from the leaf in solution Z.



[3]

Mark Allocation

^4 # f0du#frqwxrxv#qhv/#rwr#dlw#f#rg/#rwr#yhu0sslj; fh0xrvh#z d0#dv#grx0d#qhv
 [1] – fruhfw#kdsh#0r #j#xdug#fhw#e.g. *touching with rounded ends*, #r #hs#ghp d0#fhw#hlj #vtxdh#ru#
 uhfw0qj#0u,#
 ^4 #f#fk0r#s0v#w#q#j#xdug#fhw#lqg#rwr#q#hs#ghp d0#fhw; j#xdug#fhw#kdyh#k#fnhu#lqghu#z d0

- (c) Fig. 2.1 shows a diagram of a stage micrometer scale that is being used to calibrate an eyepiece graticule.

One division, on either the stage micrometer scale or the eyepiece graticule, is the distance between two adjacent lines.

The length of one division on this stage micrometer is **0.1 mm**.

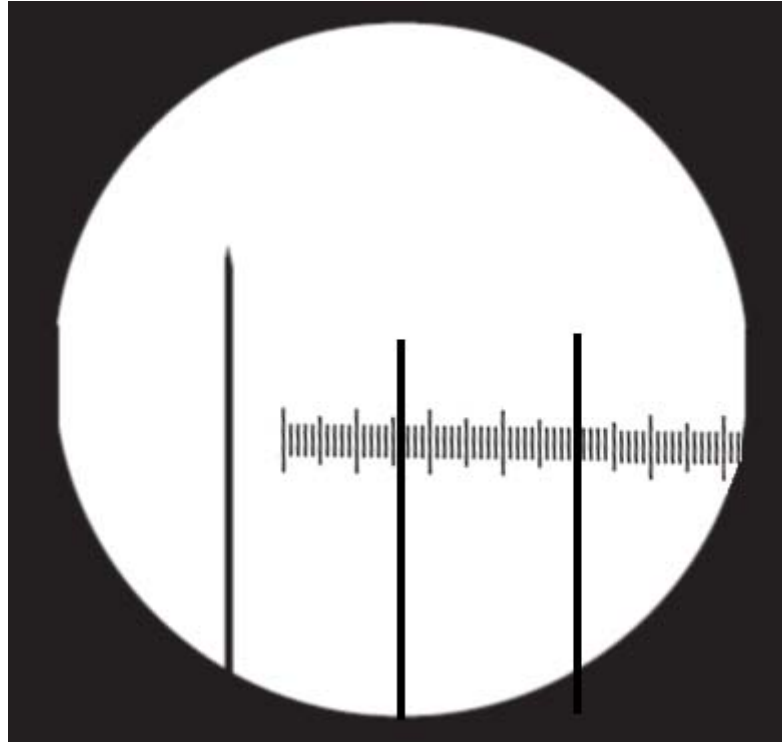


Fig. 2.1

- (i) Using this stage micrometer, where one division is 0.1 mm, calculate the actual length of one eyepiece graticule division, using Fig. 2.1.

Convert your answer to a measurement units most suitable for use in light microscopy. Show the steps and units in your calculation.

Number of eyepiece graticule divisions in 1 stage micrometer division = 24

Length of 1 stage micrometer division = 0.1 mm

Actual length of 1 eyepiece graticule division = $0.1 / 24$ mm

$$= 0.1 / 24 \times 10^3 \text{ } \mu\text{m}$$

$$= 4 \text{ } \mu\text{m} \text{ (or } 4.1 \text{ } \mu\text{m)}$$

[2]

Mark Allocation

[1] – correct calculation for 1 eyepiece graticule division, by dividing 0.1 mm by 24

[1] – correct conversion to μm , by multiplying by 1000, answer expressed to whole number AND with appropriate units

[Accept: Answer up to 1 dec. pl.]

Fig. 2.2 shows a photomicrograph of plant cells some of which have lost water by osmosis.



Fig 2.2

A student, using a prepared slide from which this photomicrograph was taken, measured the total length of the seven chloroplasts, labelled in **cell Z** in Fig 2.1.

Fig. 2.3 shows the view that the student saw when using the eyepiece graticule, calibrated in **c(i)** at the high-power of a microscope.

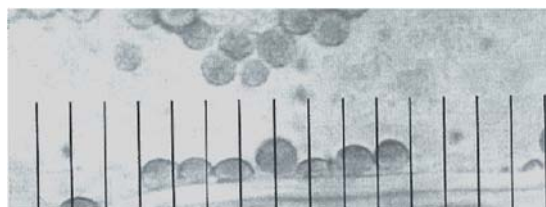


Fig 2.3

- (ii) Using this and the information in **c(i)**, calculate the actual mean length of one chloroplast as shown in Fig 2.3.

Show the steps and units in your calculation.

$$\text{Total length of 7 chloroplasts} = 8 \times 4 \mu\text{m} \quad (\text{or } 8 \times 4.1 \mu\text{m})$$

$$\begin{aligned} \text{Actual mean length of 1 chloroplast} &= (8 \times 4) / 7 \mu\text{m} \quad (\text{or } (8 \times 4.1) / 7 \mu\text{m}) \\ &= 5 \mu\text{m} \quad (\text{or } 4.6 \mu\text{m}) \end{aligned}$$

actual mean length of one chloroplast ...**5 μm**(or **4.6 μm**)..... [2]

Mark Allocation

- [1] – expressing the total length of 7 chloroplasts: using answer in **c(i)** multiplied by 8 (eyepiece graticule divisions)
- [1] – correct calculation for mean length of one chloroplast, by dividing total length (of 7 chloroplasts) by 7, expressing answer correct to whole number AND with appropriate units
[Accept: Answer up to 1 dec. pl.]

(d) Fig. 2.4 and Fig. 2.5 are photomicrographs of the lower surface of the leaf from two different plants, with the same field of view, using the same objective lens.

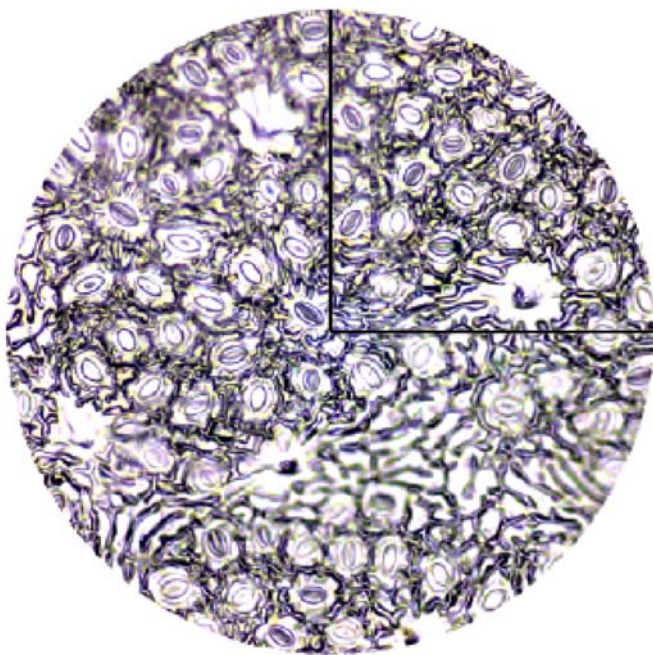


Fig. 2.4

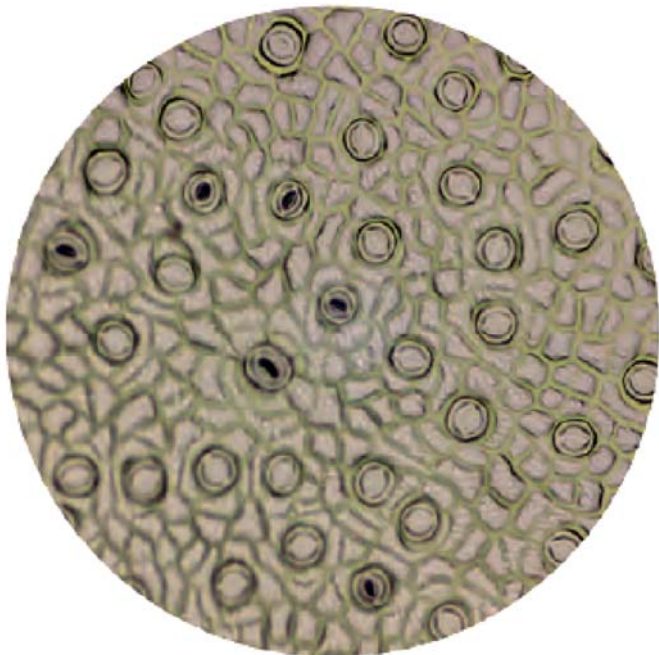


Fig. 2.5

Complete the table below to record 2 observable differences between the surface of each leaf shown in Fig. 2.4 and Fig. 2.5.

Feature	Fig. 2.4	Fig. 2.5

[2]

Mark AllocationAny **TWO** of the following:

Marking Point	Feature	Fig. 2.4	Fig. 2.5
1	Number of stomata	More (or an example of a number) closer packed / nearer each other / gap between stomata narrower / more clustered	Few(er) (or an example of a number)
2	Size of stomata / epidermal cells / guard cells	Small(er)	Large(r)
3	shape of stomata / guard cell	Oval or slit or elongated	Round(er) or circular or more curved
4	Appearance of stomata	More closed or fewer open	Less closed or more open
5	Shape of epidermal cell or pattern of lines	Very irregular or not clear Folded	Clear or anular or corners; Smoother
6	Thickness of epidermal cell walls	Thin(ner)	Thick(er)

CANDIDATE
NAME

CLASS 2T

- 3 The enzyme urease is a catalyst of the hydrolysis of urea in solution, forming ammonia and carbon dioxide, for example in the breakdown of urea in soils by microorganisms.

You are required to plan an investigation to compare the activity of urease free in solution and urease immobilised in alginate beads.

As the reaction proceeds, the ammonia released dissolves, causing the pH to increase.

You are provided with the following equipment which you may use or not in your plan, as you wish. You may **not** use any additional equipment in your plan.

- an unlimited supply of calcium alginate beads, all of uniform size, prepared with a 50 g dm^{-3} urease solution (you may call this immobilised urease)
- an unlimited volume of 50 g dm^{-3} urease solution (you may call this free urease)
- an unlimited volume of 1.0 mol dm^{-3} urea solution
- an unlimited volume of distilled water
- beakers and flasks of different sizes
- stopwatch
- broad and narrow range of pH papers and liquids with appropriate colour charts, pH probes and meters
- colorimeter and tubes/cuvettes
- thermometer
- thermostatically-controlled water baths
- graduated pipettes and pipette fillers
- filter funnels
- syringes
- glass rods for stirring
- test-tubes and boiling tubes
- test-tube racks

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- include a clear statement of the hypothesis or prediction
- identify the independent and dependent variables
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of the apparatus used
- describe the method with details and explanations of the procedures that you would adopt to ensure that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment

[Total: 14]

ANS [L3]

Step 1: Outline structure and set priority [which to be done first #1...]

Awful	H.I.V.		are	probably	still	residing	inside
[AIM]	[Hypothesis/Introduction/Variables]		[Apparatus]	[Procedure]	[Safety]	[Results]	
	[Insights/conclusn]						
#1	#2	#3	#4	#10	#5 Control #9 Procedure	#8 #6 Graph #7 Table	#11

*The above strategy saves time as ensures all essential parts of Planning are covered.

#1 AIM / HYPOTHESIS [MAX 1][Total 1]

- 1
 - 1a] Rate of hydrolysis is faster using free enzyme
 - 1b] quantity of urea hydrolysed over time is greater with free enzyme
 - 1c] immobilised urease catalyses reaction over much longer period of time

#2#3 Introduction / Theory to support hypothesis [MAX 1][Total 2]

- 2
 - Reference to enzyme active site refs to accessible active sites
 - diffusion of substrate into alginate beads
 - stability of enzyme in alginate beads

#4 Variable [MAX 2][Total 4]

3. Independent Variable
 - 3a] Concentration of Urase [Free]
 - 3b] Concentration of Urase [immobilized]/[Alginate]
4. Dependent Variable
 - 4a] pH as a measure of ammonium carbonate [or outline in procedure]
5. At least two control variables : [any 2]
 - 5a] temperature,
 - 5b] concentration of urea solution.
 - 5c] volumes used.
 - 3d] number of beads

#10 Apparatus [nil]

cite apparatus

Procedure Control [Max 1][Total 5]

#5 6. Negative control: denatured Urase

#9	Procedure Protocol	[Max 3][Total 8]
7.	method of following the reaction taking samples at intervals and calculating the initial rate	
8.	Justification/evaluation. of strategy ; e.g. can only alter concentration of immobilised enzyme by changing number of beads /limitations of colour comparison these could be awarded at the end of the plan	
9.	Method of determining. pH / (the concentration) of ammonium carbonate, at intervals ; e.g. use of pH indicator, to follow colour change	
10.	use range of concentrations of urea [Colour standard];	
11.	use range of concentrations of urease ; to find suitable concentrations to make comparison [at least 5 concentrations]	
12.	dilution table(s) included ;	
13.	method to ensure concentration of urease in reaction mixtures is the same for both free and immobilised enzyme ;	
14.	urea solution mixed with pH indicator ;	
15.	equilibration in water bath ;	
16.	mixing. urease/beads. and urea solution at time = 0 ;	
17.	staggered start ;	
18.	samples taken at stated intervals ;	
19.	repeats/replicates (calculate average in table)	
#8	Safety	[Max 1][Total 9]
20.	ref to hazards and precautions [<i>may be taken from a diagram or a flow or sequence diagram</i>] 10a] Reagents (irritants) use of Gloves 10b] Glassware fragile handle with care	
#6	Results Graph	[Max 2][Total 11]
21.	Line of best fit : plot results on appropriate graph (bar or line) ;	
22.	takes gradient to give rate of each Urase [Free / Immobilized];	
#7	Results Table	[Max 2][Total 13]
23.	Headings with Units / no units in table / data present	
24.	Replicates / repeats	
25.	Time taken (t) to reach colour standard recorded ;	
26.	rate = $1/t$; $A \ 1000/t$. etc.	
27.	colour standard set up at known pH ;	
28.	colour change followed in colorimeter ;	
#11	Conclusion	[Max 1][Total 14]
29.	calculate, standard deviation/standard error ;	

- 30. ref to use of t-test to see if rates are significantly different ;
- 31. uncertainty/precision. of results ;